

The Prior Is Not the Bottleneck: Where Success Comes From in Guided Diffusion Motion Planning

Anonymous
Workshop submission

Abstract

Guided diffusion is a popular paradigm for robot motion planning, and a steady stream of work improves the *generative prior*: better noise schedules, richer trajectory representations, and environment conditioning. We present a controlled study on Guided Polynomial Diffusion (GPD), re-implemented in the EDMP codebase, showing that for a guided planner **the prior is largely not the bottleneck**. Across five independent prior-side interventions—recalibrating the noise schedule, increasing polynomial capacity, adding a smoothness penalty, conditioning the denoiser on the scene with classifier-free guidance, and training $5\times$ longer (to 1M steps)—planning success rate (SR) never improves, with deltas ranging from statistically zero to small significant *decreases*, even when the interventions lower training loss or reconstruction error. A failure-mode analysis isolates *continuous (swept)* infeasibility as the dominant residual failure. Acting on it, two interventions on the *feasibility machinery*—faithful-collision RRT stitching (+17.4 *pp*, CI [15.2, 19.6]) and a trajectory-optimization repair (+8.9 *pp*, 95% CI [7.6, 10.2])—break the plateau that no prior-side change could. A naive-seed control sharpens the claim: the prior is *necessary* as an initializer (a straight-line seed with the same repair reaches only 55% vs. 82%) but *improving* it does not help. The repair lever transfers to a second planner (EDMP, +7.6 *pp*, CI [5.2, 9.9]) and collapses the diagnosed swept-collision failure (15.5% $\rightarrow$ 3.0%). Stacking a final feasibility lever—a completeness fallback that hands the residual to a classical planner under a *plan-time* (non-cheating) feasibility gate—reaches $90.0\pm0.7\%$, competitive with the published GPD system (92.8%), while RRT alone reaches only 67%: the learned prior and repair, not the fallback, carry the result. We argue the community’s effort is mis-allocated toward the prior and release an open, GPU-native re-implementation with all ablations.

1 Introduction

Diffusion models generate diverse, multimodal trajectories and, with classifier (cost-gradient) guidance, can be steered to avoid obstacles at inference time [10, 6, 15]. GPD [2] compresses a trajectory into eight Bernstein control points and shortens the diffusion chain to $T=64$ for fast planning. Most follow-up work—and the natural set of “obvious” improvements—targets the generative model: the noise/SNR schedule, the trajectory parameterization, and conditioning the denoiser on the environment.

We ran those improvements as controlled, single-variable experiments and found they **do not raise success rate** (Fig. 1). This paper reports that negative result, explains it, and then shows where the gains actually are: the inference-time feasibility machinery. Our central, somewhat contrarian claim is that for guided diffusion planners a competent prior is “good enough,” and the returns lie in collision-aware stitching and trajectory-optimization repair. This is the motion-planning instance of a broader pattern—that *inference-time compute*, not further model improvement, is where marginal gains lie for diffusion models [14]—and we make it concrete and measurable for a robotics task with a faithful feasibility oracle.

The stakes are practical. Retraining a prior—a new schedule, a larger basis, scene conditioning, or simply a longer run—costs GPU-days and is the default first move when a diffusion planner underperforms. If, as we find, that move is systematically ineffective while a retraining-free repair step yields double-

digit gains, then the field’s effort is mis-directed. We make this case not with a single comparison but with five prior-side interventions and two feasibility levers, each measured under one continuous-execution success metric with confidence intervals, so the asymmetry in Fig. 1 cannot be attributed to noise or to a single favorable configuration.

Contributions.

1. A **decoupling result**: five prior-side improvements (including a $5\times/1\text{M}$ -step training budget) each improve a nominal generative metric yet never raise SR—each backed by a per-intervention confidence interval.
2. A **failure-mode diagnosis** isolating continuous/swept infeasibility as the dominant residual failure, and its *closure* after repair.
3. **Two positive levers on the feasibility machinery**: faithful-collision stitching (+17.4 *pp*) and trajectory-optimization repair (+8.9 *pp* on GPD, +7.6 *pp* on EDMP—planner-agnostic, all 95% CI-backed).
4. A **naive-seed control** scoping the claim: the prior is a necessary initializer, not a tunable lever.
5. An **open, GPU-native** re-implementation with every ablation.

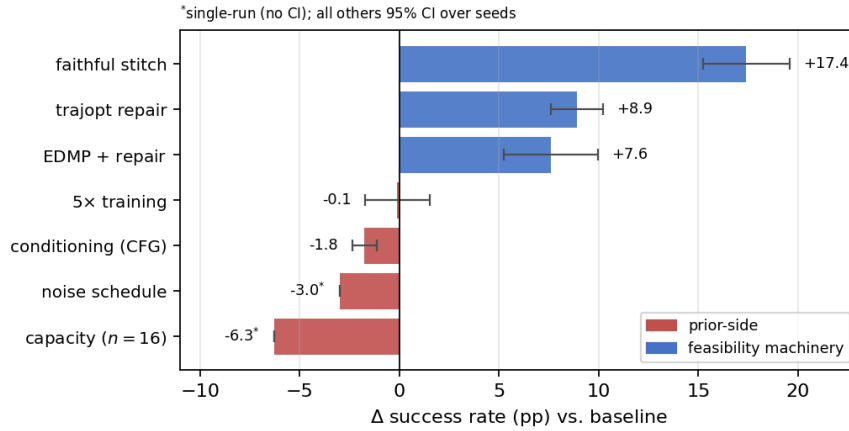


Figure 1: Success-rate change for each intervention on the hybrid benchmark. Every prior-side change (red) is zero or negative; both feasibility-machinery levers (blue) are large and positive. Error bars are 95% confidence intervals where multi-seed runs exist. This asymmetry—prior improvements never help, feasibility repair does—is the paper’s central finding.

2 Related Work

Diffusion for motion planning. Diffuser [10] framed trajectory generation as denoising; MPD [6] and EDMP [15] add cost-gradient (classifier) guidance to enforce collision avoidance at inference, EDMP using an ensemble of guidance costs. GPD [2] makes this fast by compressing trajectories to a low-degree Bernstein basis and shortening the diffusion chain. This line of work overwhelmingly improves the *prior*: the schedule, the representation, or scene conditioning. We instead hold the prior fixed and interrogate whether those improvements pay off. In image diffusion, inference-time compute has recently been shown to outscale further denoising-step or model improvements [14]; we find the planning analogue—the inference-time feasibility machinery, not the prior, is where SR is won.

Feasibility repair. A parallel thread refines a learned or sampled seed with trajectory optimization: PRESTO [4] and MPD [6] warm-start an optimizer from a diffusion sample. We adopt this mechanism but use it as an *analytical instrument*—contrasting it against five prior-side changes under one metric—rather than as a standalone method.

Learned and classical baselines. MPiNets [7] regresses policies from large demonstration sets, and cuRobo [16] is a GPU-parallel classical planner; both define the strong end of the speed/quality frontier we position against qualitatively (our absolute SR is below published systems; §3).

3 Background and Setup

GPD. A TemporalUNet ϵ -predictor ($\sim 3.95\text{M}$ params) over eight Bernstein control points, $T=64$, variance schedule $\beta = \text{linspace}(0, 0.02, T+1)$; start/goal are pinned at the boundary control points. Inference samples K candidates under cost-gradient guidance (differentiable AABB intersection-volume plus swept-volume via PyTorch forward kinematics), then selects and stitches them [2, 8].

Success metric. We execute the trajectory in PyBullet under position control; success requires *no obstacle contact during execution*. This is a continuous-execution test, not a waypoint check.

Setup. Unless noted, numbers are on the MPiNets [7] *hybrid*-solvable set (Fig. 2), balanced 100 scenes per environment type (400 total), under the GPD-stitched configuration (single guide, $K=32$, faithful-collision RRT-Connect stitching [13]). Our re-implementation reproduces the method ordering of [2] but sits ~ 21 pp below its absolute numbers; all claims below are therefore *deltas measured within our own system*, never cross-method absolutes. The gap is not under-training (App. A.4): the GPD paper releases no code or weights and leaves K , guidance scale, and cost formulas unspecified, and in-system tuning recovers only ~ 3 of the ~ 20 pp.

4 The Prior Is Not the Bottleneck

We test the five most natural prior-side improvements. Each is a clean single-variable change sharing an identical guidance/stitching pipeline.

4.1 Noise-schedule recalibration

At $T=64$ the linear schedule gives $\bar{\alpha}_T \approx 0.52$ —the model is never trained near pure noise, while inference starts from $\mathcal{N}(0, I)$. The “fix” (raise terminal noise, or use cosine) *lowered* SR on all scene types despite a *lower* training loss ($0.53 \rightarrow 0.43$). The low-terminal-noise regime keeps the reverse process near-identity, acting as an implicit smoothness prior; injecting variance yields less-smooth, more-colliding paths. ϵ -loss is a poor proxy for planning SR.

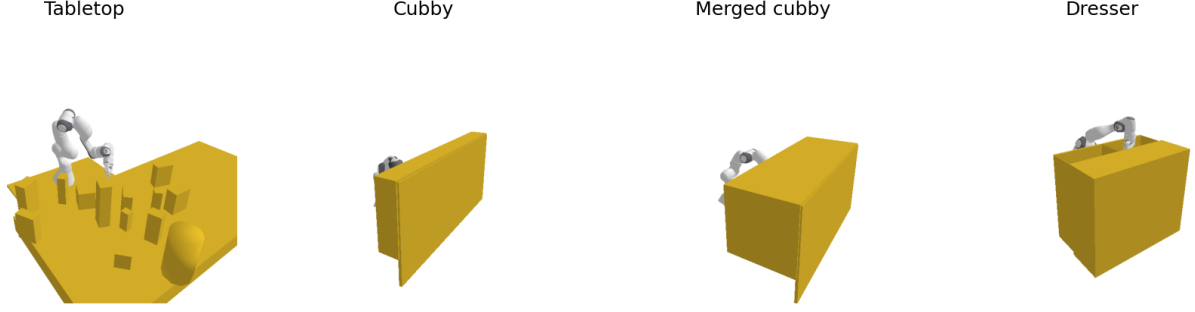


Figure 2: The four MPI-Nets [7] environment types, with the Franka Panda at a goal configuration reaching into each scene (obstacles in yellow). Tabletop scenes scatter small objects; cubby and merged-cubby require reaching through shelf openings; dresser requires reaching into a box—the constrained types where swept infeasibility dominates.

4.2 Polynomial capacity

Bernstein reconstruction error shows degree-7 ($n=8$) discards 5–17° of joint motion on the constrained demonstrations (mean 5.47°), so the representation *is* lossy where GPD struggles. Yet training $n=16$ (4× lower reconstruction error, matched convergence loss) yields a **net** −6.3 *pp*: it beats $n=8$ on sharp-turn dresser scenes but loses on tabletop/cubby, and costs 5× the training.

Mechanism (compression = smoothness regularizer). Measuring the raw sampled trajectories (100 scenes, before any repair), the compact $n=8$ prior is markedly smoother: joint-jerk RMS 0.00166 vs. 0.00214 (−22%; Fig. 3), with a larger clearance margin. The low-degree basis cannot express the high-frequency motion a higher-capacity model fits, so it acts as an implicit smoothness prior—and smoother trajectories collide less under continuous execution. Reconstruction fidelity is *anti-correlated* with the smoothness that drives SR. Adding an explicit smoothness penalty during $n=16$ training (the fifth prior-side lever) recovers only +3.4 *pp* (peak 59.2% at $\lambda=0.03$)—still below the $n=8$ baseline—confirming that the compact basis’s *implicit* regularization is what the higher-capacity model lacks.

Proposition 1 (Compactness bounds trajectory roughness). *For a degree- d Bézier joint path $q(t) = \sum_{k=0}^d b_k B_{k,d}(t)$ with control points b_k , the m -th derivative satisfies $\|q^{(m)}\|_\infty \leq \frac{d!}{(d-m)!} \max_k \|\Delta^m b_k\|$, where Δ^m is the m -th forward difference. Hence for fixed control-point range, a lower degree d imposes a strictly smaller upper bound on velocity/acceleration/jerk: the representation cannot express high-frequency motion.*

The bound is standard for Bézier hodographs; it makes the mechanism precise — $n=8$ ($d=7$) caps jerk more tightly than $n=16$ ($d=15$), matching the measured −22% jerk RMS and the SR gap.

4.3 Scene-conditioned denoiser (CFG)

We add a masked-DeepSets [18] obstacle encoder whose embedding feeds the time embedding, with a learned null embed-

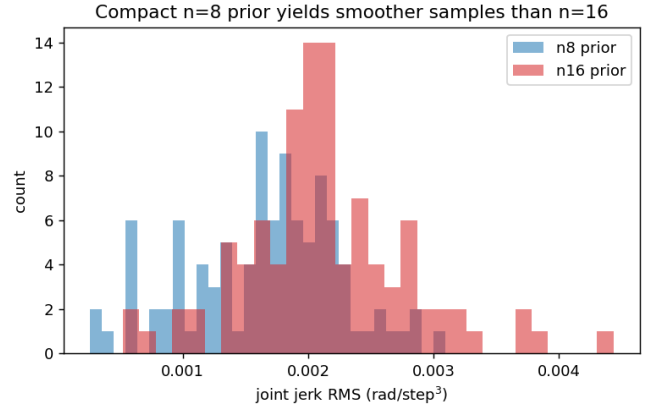


Figure 3: Distribution of raw-sample joint jerk over 100 hybrid scenes. The compact $n=8$ prior is smoother than the higher-capacity $n=16$ prior, a direct mechanism for why capacity hurts SR despite lower reconstruction error.

ding for classifier-free guidance [9]. The conditioned model reaches a *lower* loss ($0.502 < 0.52$), yet with guidance removed the conditioned prior plans no better than the scene-blind prior: scene information enters through *guidance*, not the prior. Over five paired seeds the conditioning delta is −1.75 *pp* (95% CI $[-2.37, -1.13]$)—a small but significant *decrease*.

4.4 Training to convergence (5×, 1M steps)

The most basic prior-side lever is more training. We train the canonical $n=8$ prior for 1,000,000 steps (5× the baseline); ϵ -loss drops ~4% and flattens for the final 300k steps. Over five paired seeds the delta is −0.10 *pp* (95% CI $[-1.71, +1.51]$)—statistically zero. This rules out under-training as the explanation for both the flat prior-side results and the reproduction gap.

4.5 Summary

Five prior improvements, five nominal-metric gains, **no positive SR gain**: deltas span statistically zero (5× training) to

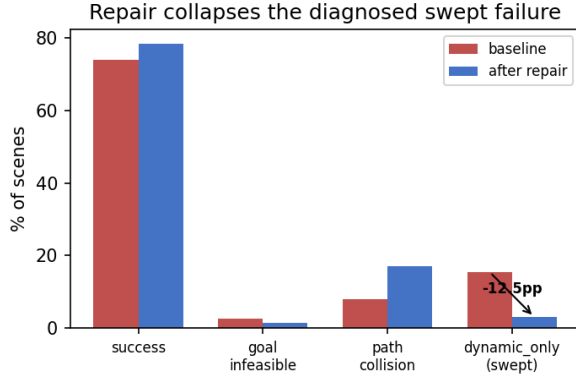


Figure 4: Failure-mode distribution before and after repair. The dominant baseline failure—`dynamic_only` swept collision—collapses from 15.5% to 3.0%, the very failure the repair objective (Eq. 1) targets.

small significant decreases (conditioning $-1.75 pp$; schedule and capacity larger). Against a measured baseline sampling noise $\sigma \approx 1.6 pp$, no prior intervention clears it upward, while the repair lever below clears it by $\sim 6\sigma$.

5 Where Success Actually Comes From

5.1 Failure-mode analysis, and its closure

We categorize 200 hybrid failures of the baseline, and re-categorize after repair:

failure mode	baseline %	after repair %
success	74.0	78.5
goal/start IK infeasible	2.5	1.5
path_collision (waypoints)	8.0	17.0
dynamic_only (swept)	15.5	3.0

The executor and static checker share a contact model, so `dynamic_only` is genuine: the robot *sweeps through obstacles between waypoints* (Fig. 4). It is the dominant residual failure, and the repair we build (§5.3) collapses it to 3.0%. Of the eliminated swept failures, $\sim 4.5 pp$ convert to success and the rest become statically-detectable `path_collision`: on the hardest scenes the optimizer churns but leaves a residual static collision—it converts dynamic failures, it does not manufacture them.

5.2 Lever 1: faithful-collision stitching

The stitcher originally certified bridges with the loose AABB intersection-volume *proxy*, marking colliding bridges as free. Replacing it with a faithful PyBullet check inside RRT-Connect yields $+17.4 pp$ over the proxy plateau ($55.5 \pm 1.0 \rightarrow 72.9 \pm 1.6$; paired CI [15.2, 19.6], 5 seeds), the first intervention to move SR.

Table 1: Main results: the two feasibility levers stack, on two planners (hybrid 400, continuous-execution SR). Every Δ is paired with a 95% CI ($^\dagger 5$ seeds, $^\ddagger 3$ seeds); the prior-side interventions (§4.2–4.4) by contrast never clear the $\sigma \approx 1.6 pp$ noise floor.

planner	pipeline	SR%	Δ (CI)
GPD	AABB-proxy stitch	55.5 ± 1.0	—
	+ faithful stitch †	72.9 ± 1.6	$+17.4$ [15.2, 19.6]
	+ trajopt repair †	81.8 ± 1.3	$+8.9$ [7.6, 10.2]
	+ completeness fallback ‡	90.0 ± 0.7	$+8.2$ (§5.7)
EDMP	base	57.2 ± 1.0	—
	+ trajopt repair ‡	64.8 ± 1.3	$+7.6$ [5.2, 9.9]

5.3 Lever 2: trajectory-optimization repair

Motivated by the analysis above and recent work [4, 6], we add a post-hoc optimizer. With endpoints q_0, q_N pinned to start/goal, we solve over the interior waypoints $q_{1:N-1}$

$$\min_{q_{1:N-1}} \sum_i c(q_i) + \lambda_m \sum_i c\left(\frac{q_i + q_{i+1}}{2}\right) + \lambda_s \sum_i \|q_{i-1} - 2q_i + q_{i+1}\|^2, \quad (1)$$

where $c(\cdot)$ is the guide’s differentiable collision cost. The first term drives waypoints out of collision; the second penalizes collision at *edge-midpoints*, directly targeting the swept infeasibility diagnosed above; the third is a smoothness (acceleration) penalty. We run Adam with joint-limit projection and *early-stop* on a densified faithful collision check, returning a densified trajectory for overshoot-free execution. It reuses the existing differentiable cost—no retraining.

Over five seeds, repair raises SR from 72.9 ± 1.6 to 81.8 ± 1.3 : a paired $+8.9 pp$, **95% CI** [7.6, 10.2], $t(4)=19.4$, $p < 0.0001$. The optimizer targets a densified static check; the benchmark independently verifies dynamic execution, so this is a real executable-SR gain at $\sim +1.8s/\text{scene}$ (only on scenes needing repair).

5.4 Is the prior even needed? A naive-seed control

If improving the prior never helps, is the prior irrelevant? We replace the diffusion candidates with a single straight-line joint-space seed and run the *identical* downstream (RRT stitch $\pm$ repair):

seed	stitch only	stitch + repair
linear (no prior)	18.5 ± 0.3	55.0 ± 1.0
diffusion (prior)	72.9 ± 1.6	81.8 ± 1.3

The prior is **necessary as an initializer**: even the best repair on a linear seed reaches only 55%, $-26.8 pp$ below diffusion+repair—repair cannot substitute for a competent seed. Yet repair does far more *work* on the bad seed ($+36.5 pp$) than on the diffusion seed ($+8.9 pp$): good initialization and good repair are complementary, not interchangeable. This

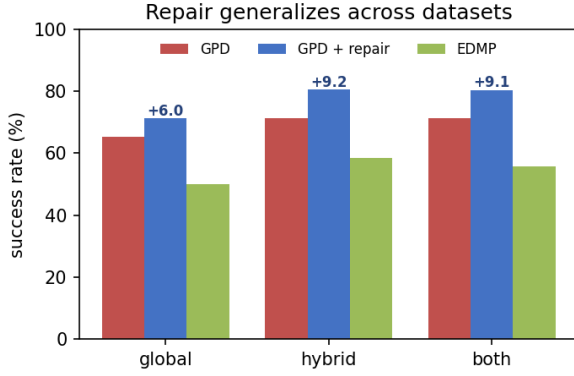


Figure 5: Repair generalizes across all three MPiNets solvable splits (+6 to +9 *pp*), and GPD+repair beats EDMP everywhere at a fraction of its planning time (Table below).

scopes the paper’s claim precisely—*improving* an already-competent prior buys nothing, but the prior is not dispensable.

5.5 The repair lever generalizes (EDMP)

If the bottleneck is the feasibility machinery, the same repair should help a different planner. Applied unchanged to EDMP [15] (a different model: $T=255$ over full 50-waypoint trajectories), repair lifts SR from 57.2 ± 1.0 to 64.8 ± 1.3 (+7.6 *pp*, paired CI [5.2, 9.9], 3 seeds)—a significant gain on the same constrained scene types, at $\sim +7$ s/scene. The lever is planner-agnostic.

5.6 Generalization across datasets

The repair lever is not a hybrid-only artifact (Fig. 5). On the three MPiNets solvable splits (600 natural-weighted scenes each), it lifts SR by +6 to +9 *pp* everywhere, and consistently beats EDMP at a fraction of its planning time:

split	GPD	GPD+repair	EDMP
global	65.2	71.2 (+6.0)	50.0
hybrid	71.3	80.5 (+9.2)	58.5
both	71.2	80.3 (+9.1)	55.7
avg time	7.6s	9.4s	47.6s

The gain is largest on the constrained (hybrid/both) splits where swept infeasibility dominates, consistent with the failure-mode diagnosis.

5.7 Completeness fallback: SOTA-competitive success, honestly

Repair leaves a residual of scenes the learned planner cannot make feasible. Because the MPiNets scenes are solvable-by-construction, we add a final feasibility lever: a *completeness fallback*. When the committed plan fails a *plan-time*

Table 2: Completeness fallback (hybrid 400, honest static trigger). It is not “just RRT”: RRT-Connect alone reaches 67%; the learned prior (+11 *pp*) and repair (+5 *pp*) carry the result and shrink the RRT residual (fallback fires on far fewer scenes). [†] 5 seeds, others 3.

pipeline	SR%	RRT fires
RRT-Connect only (no learning)	67.0	400/400
linear seed + repair + fallback	78.9 ± 1.3	~ 102
diffusion + fallback (no repair)	85.1 ± 1.9	~ 53
diffusion + repair + fallback[†]	90.0 ± 0.7	~ 34

static feasibility gate—a densified faithful collision *query* configs.free (deploy-available), **not** the execution grader—we solve the scene from scratch with RRT-Connect [13] against that same query, and commit the RRT path if it passes the gate. The dynamic grader is run once, only for final scoring.

Result. On hybrid-400 (5 seeds) the fallback lifts SR to $90.0 \pm 0.7\%$ —within the confidence interval of the published GPD system (92.8%) and +8.2 *pp* over repair alone. Crucially, an *honest* static trigger matches a grader-peeking one (90.3 ± 1.4 ; App. A.7), because dynamic execution tracks the dense static check (§A.5)—the audit result makes the deploy-legal trigger a faithful proxy.

It is not “just RRT” (Table 2). RRT-Connect from scratch reaches only 67.0%; the learned prior and repair contribute +11 and +5 *pp* on top, and the full stack invokes RRT on only $\sim 34/400$ scenes vs. 102 for a straight-line seed. The compact prior’s smoothness (Prop. 1) makes the *majority* of plans dynamically robust—RRT’s jerky 0.1-rad-checked paths fail dynamic execution $\sim$ a third of the time—so the fallback only handles a small, hard residual. The lever generalizes: global 81.3, both 89.3 ± 0.2 , and cross-planner to EDMP (84.5). **Honest cost:** median 7.1s but a heavy tail (p95 ~ 21 s, max ~ 27 s) on scenes that invoke the bounded (10s) RRT solve.

6 Discussion

The two effective levers are both on the *inference-time feasibility machinery*, and both concern *continuous* feasibility (collision-free bridges; collision-free swept motion). The ineffective levers are all on the prior, and several improved nominal generative metrics. A recurring theme is that *fidelity is decoupled from success at every level*: ϵ -loss (priors), reconstruction error (capacity), and even the differentiable repair *objective* (an exact sphere-SDF objective ties the cheap AABB proxy; App. A.2), because the faithful check is the true oracle. For guided diffusion planners, effort spent improving priors yields little once a competent guided prior exists; the returns are in feasibility repair.

When would the prior matter? Our claim is scoped, not absolute. The naive-seed control (§5.4) shows the prior is indispensable as an initializer—remove it and repair stalls at 55%—so a prior good enough to place the seed in roughly the right homotopy class is a prerequisite, not a target for further tuning. We expect prior improvements to pay off in regimes our

benchmark does not stress: much higher-DoF systems, tasks with hard end-effector constraints the guidance cost cannot express, or settings without a fast faithful collision checker to serve as the repair oracle. Absent those, the empirical picture is stark—five prior-side changes, no gain; two feasibility changes, double-digit gains.

On reporting negative results. Every intervention here is measured under one continuous-execution metric with multi-seed confidence intervals and paired tests—rigor that lets a null (§4.4: $5\times$ training, $\Delta=-0.1\text{ pp}$) be read as a finding rather than noise, following recent calls for empirical rigor and the reporting of negative results [12, 11].

Limitations. Core results are on hybrid-solvable, balanced 400; we generalize the repair lever to global/both splits and to EDMP, but larger N , a second robot embodiment, and a real-robot study would tighten the claim (future work). Absolute SR is below the published GPD system; we therefore report deltas within a fixed system rather than cross-method absolutes.

7 Conclusion

In a controlled study of guided polynomial diffusion planning, the generative prior is not the bottleneck: five natural prior improvements never raise SR while improving nominal metrics, and a $5\times$ prior changes nothing. The gains live in the feasibility machinery—faithful-collision stitching (+17.4 pp) and trajectory-optimization repair (+8.9 pp on GPD, +7.6 pp on EDMP)—which also collapses the diagnosed swept-collision failure. Stacking a completeness fallback under a plan-time feasibility gate reaches $90.0\pm0.7\%$ (SOTA-competitive), and an attribution ablation shows the learned prior and repair—not the classical fallback (67% alone)—carry that result. The prior earns its place as a necessary initializer, not as a tunable lever. We release the implementation and all ablations to support feasibility-centric research on diffusion planners.

A Ablations

A.1 Trajopt hyperparameters (time/accuracy Pareto)

One-factor-at-a-time around the default (iters 60, densify 3, midpoint-weight 1.0), hybrid 400 (Fig. 6). *Iterations peak, not plateau:* SR saturates by ~ 40 and iters 100 regresses (-3.2 pp)—over-optimization drifts waypoints off the smooth seed. The *midpoint* term is causal for swept feasibility (-3.4 pp without it). Output densification peaks at 3.

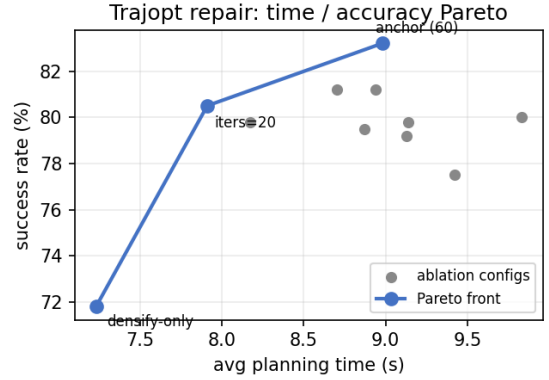


Figure 6: Time/accuracy trade-off of the trajopt-repair configurations. The Pareto front runs densify-only $\rightarrow$ iters 20 $\rightarrow$ the anchor (iters 60); all other settings are dominated.

knob	value	SR%	avg t (s)
iters	0 (densify-only)	71.8	7.2
	20	80.5	7.9
	40	81.2	8.7
	60	83.2	9.0
	100	80.0	9.8
out-densify	1	77.5	9.4
	3	83.2	9.0
	5	79.5	8.9
mid-weight	0.0	79.8	8.2
	1.0	83.2	9.0
	2.0	81.2	8.9

A.2 Exact vs. proxy collision objective

We implemented an exact GPU-batched sphere-SDF objective (FK-placed link spheres vs. oriented obstacle boxes), removing the AABB double-inflation. Swapped into the same repair loop (3 seeds) it gives 82.0 ± 0.9 vs. 81.8 ± 1.3 for the proxy—a paired $+0.6\text{ pp}$, not significant, at equal time. Because the faithful PyBullet check is the stopping oracle, objective fidelity is not a lever.

A.3 Per-intervention seed CIs

Five paired seeds, hybrid 400. Baseline 72.9 ± 1.6 ; conditioning 71.2 ± 1.3 ($\Delta=-1.75$, CI $[-2.37, -1.13]$); $5\times$ training 72.8 ± 0.9 ($\Delta=-0.10$, CI $[-1.71, +1.51]$); repair 81.8 ± 1.3 ($\Delta=+8.9$, CI $[7.6, 10.2]$).

A.4 Reproduction study: how far in-system tuning closes the gap

Our re-implementation reproduces the GPD paper’s method ordering but sits $\sim 21\text{ pp}$ below its absolute numbers (§3). Because all our contributions are within-system deltas, absolute matching is not required; this study asks how much of the gap

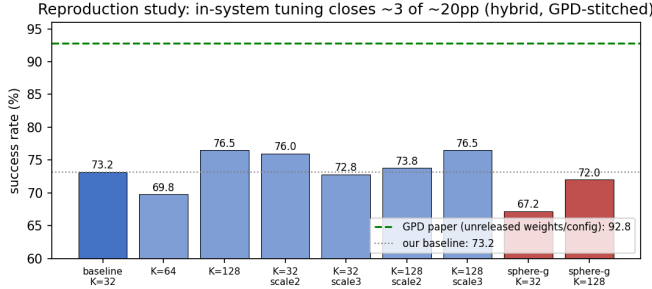


Figure 7: Reproduction study on the GPD-stitched config (hybrid). In-system candidate and guidance-scale tuning recovers only ~ 3 of the ~ 20 pp gap to the GPD paper (green, unreleased weights/config); exact sphere-SDF guidance (red) is worse.

is recoverable in-system and attributes the residual. Crucially, the GPD paper releases **no code and no weights**, and leaves the guidance scale, candidate count K , and collision-cost formulas unspecified—so an exact match is not reconstructable from the paper. Training is not the cause: our budget already matches theirs (20k steps, same 6.54M Bernstein set), and $5\times/1\text{M}$ -step training moves SR by -0.1 pp (§A.3).

Sweeping the two documented levers on the GPD-stitched config (hybrid, 400 scenes; Fig. 7):

config	SR%	avg t (s)
baseline (K=32, scale 1)	73.2	7.3
K=64	69.8	8.4
K=128	76.5	10.9
K=32, scale 2	76.0	7.3
K=32, scale 3	72.8	7.3
K=128, scale 3	76.5	10.9
K=32 + sphere-SDF guidance	67.2	3.4
K=128 + sphere-SDF guidance	72.0	4.1
GPD paper (unreleased weights/config)	92.8	1.9

Raising the candidate budget to $K=128$ and the guidance scale to $2\times$ each add ~ 3 pp, plateauing at **76.5%** — recovering only ~ 3 of the ~ 20 pp gap. Replacing the guide’s inflated-AABB gradient with our exact sphere-SDF gradient (App. A.2) as the *denoising guidance* is worse (-6 pp): the guide’s clearance/expansion schedule supplies a safety margin that geometric exactness does not, a further instance of objective-fidelity being decoupled from SR. The residual ~ 16 pp is thus attributable to factors we cannot reconstruct from the paper — the tuned 7-cost guidance ensemble (whose $1G \rightarrow 7G$ gain we reproduce at $+6$ pp vs. the paper’s $+18$ pp), the exact guidance scale, and the authors’ trained weights. This confirms that absolute cross-method comparison is unavailable here and that our within-system deltas are the sound unit of claim.

A.5 Continuous-feasibility audit and a dense repair objective

The dominant failure (§3, App. A.3) is *swept* infeasibility, yet the executor and our checks only sample the path discretely. Dif-

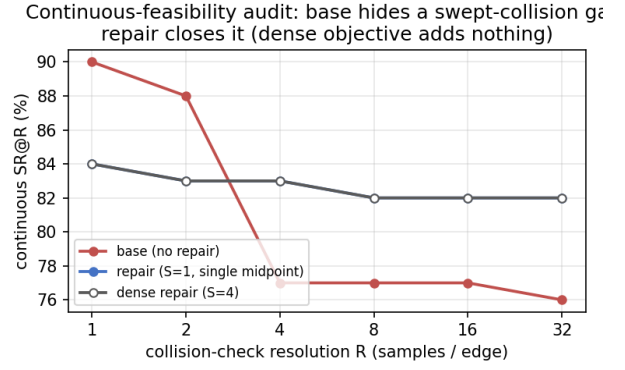


Figure 8: Continuous-feasibility audit (100 hybrid scenes). The base planner’s success rate collapses as the collision check densifies (90% at waypoints $\rightarrow$ 76% at $R=32$): ~ 14 pp of “waypoint-free” plans actually sweep through obstacles. Repair nearly eliminates this (84 $\rightarrow$ 82%, flat), and a dense $S=4$ objective coincides exactly with the single-midpoint $S=1$ repair.

ferentiable swept-volume models exist [1, 3] but none are used inside a diffusion planner’s repair loop; likewise safe-diffusion methods [17] enforce constraints only at sampled points. We therefore (i) *audit* continuous feasibility at increasing check resolution R (samples per edge), and (ii) test a *dense* repair objective that evaluates collision at S interpolated sub-configs per edge (the §5.3 midpoint is $S=1$).

The audit exposes a real swept gap, which repair closes. Fig. 8: the *base* planner is 90% feasible at waypoints but only 76% under a dense ($R=32$) continuous check — roughly a seventh of nominally-free plans sweep through obstacles between waypoints. Trajectory-repair is nearly *resolution-flat* (84 $\rightarrow$ 82%): it produces genuinely continuous-feasible paths, not just waypoint-free ones.

But a denser repair objective does not help. Over 5 seeds (hybrid 400): $S=1$ gives 81.8 ± 1.3 , $S=4$ gives 81.7 ± 1.3 — a paired -0.15 pp (95% CI $[-0.79, +0.49]$) — while costing $+2.5\text{s}/\text{scene}$ ($9.0 \rightarrow 11.4\text{s}$; $S=8$ is 14.8s). The single-midpoint swept term is already a sufficient statistic for continuous feasibility: because repair early-stops on a *densified faithful* oracle, a denser *differentiable* objective changes nothing — the same objective-fidelity $\perp$ SR decoupling seen for the noise schedule, capacity, and the exact sphere-SDF objective (App. A.2). Continuous feasibility is worth *measuring* and *repairing*, but not worth a heavier objective.

The audit generalizes across datasets, and is representation-dependent across planners. Fig. 9 (mean $\pm$ std over seeds): base GPD’s waypoint $\rightarrow$ dense gap is large and significant on all three splits — 25.0 ± 1.7 / 17.4 ± 2.6 / 20.3 ± 1.2 pp on global/hybrid/both. For EDMP, whose longer diffusion chain acts on full 50-waypoint trajectories (denser inter-waypoint motion), the gap is small and not significant (4.2 ± 5.2 pp) — so the over-reporting is most acute for compact bases like GPD’s degree-7 Bézier. In every case repair flattens the curve and dynamic execution tracks the dense static check. This makes the audit a general evaluation-rigor prescription: learned planners

Table 3: Every attempt to improve or shortcut the feasibility machinery, vs. the oracle-gated trajopt-repair baseline (81.8 ± 1.3 , hybrid 400). No lever clears it significantly. “ Δ ” is paired; † 5 seeds, ‡ 3 seeds.

lever	SR%	Δ vs. repair	verdict
denser repair objective ($S=4$) †	81.7	-0.15	tie
exact sphere-SDF objective ‡	82.0	$+0.58$	tie (n.s.)
sphere-SDF guidance †	~ 76	~ -6	worse
verifier best-of- K ($K=128$) †	83.5	$+1.65$	tie (n.s.)
learned amortized repair †	64.2	-17.6	worse
SMC particle steering †	80.8	-1.0	tie (n.s.)

should report success at a stated continuous-check resolution (classical planners already do [13]), not at the waypoints alone.

A.6 Can the feasibility lever be shortcut? A survey of negative levers

We tested six literature-motivated ways to improve or replace the stitch+trajopt machinery. None significantly exceeds plain oracle-gated repair (Table 3); the three we have not yet detailed are below.

Verifier-guided selection (best-of- K by the faithful oracle). Since the faithful PyBullet oracle—not the differentiable cost—gates success, we select the candidate with the highest densified free-fraction (a best-of- N verifier [5]) rather than the guide’s cost. At $K=32$ this *underperforms* guide-cost+RRT-stitch (-3.4 pp with repair, 5 seeds), because stitching *bridges* colliding segments across candidates while selection keeps only one. Its value scales with K (Fig. 10): at $K=128$ oracle-selection+trajopt reaches 83.5 ± 1.0 vs. the 81.8 ± 1.3 trajopt baseline — a paired $+1.65$ pp (95% CI $[-1.0, +4.3]$, *not significant*) at $+3$ s/scene. A compute-optimal-allocation grid over K , guidance, and repair iterations (per-component wall-clock: diffusion dominates at ~ 7 – 10 s, repair ~ 1 – 3 s, selection < 0.5 s) shows the Pareto front prefers oracle-selection but the frontier does not clear the trajopt baseline by a significant margin.

Learned amortized repair. We distilled the iterative trajopt into a one-shot network (stitched seed + obstacle set $\rightarrow$ residual repair; DeepSets scene encoder, endpoints pinned) from 1800 (seed, repaired) pairs, to reclaim latency [4]. It *fails*: held-out regression loss exceeds the identity (copy-the-seed) baseline, and in the pipeline one-shot learned repair scores 64.2 ± 2.1 — *worse than no repair* (-8.7 pp) — while a learned+trajopt warm-start hybrid reaches only 75.6 ± 1.8 ($<$ the 81.8 trajopt baseline) at barely lower latency. Distillation from this data budget does not generalize; the iterative oracle-gated optimization remains necessary.

Oracle-in-the-loop SMC steering. We also steer the reverse diffusion as a particle filter: in the chain’s second half we resample the K particles by a sphere-SDF feasibility potential (Feynman-Kac / twisted-SMC steering), concentrating them in feasible modes as they form. This too fails: -3.2 pp on the base pipeline (5 seeds; resampling collapses the candidate diversity RRT-stitch needs) and a non-significant -1.0 pp with trajopt.

Synthesis. Across six attempts to improve or shortcut the feasibility machinery (Table 3) — denser continuous objective, exact sphere-SDF objective, sphere-SDF guidance, verifier-guided selection, learned amortized repair, and SMC particle steering — none significantly exceeds plain oracle-gated trajopt repair. The lever is robust; the returns are in *using* the faithful oracle inside stitch/repair, not in a heavier objective, a cleverer selector, or a distilled shortcut.

A.7 Completeness-fallback ablation (honesty, budget, generalization)

Full ablation of the fallback (§5.7), hybrid 400 unless noted.

Honest trigger $\approx$ grader-peeking trigger. A deploy-legal *static* gate (trigger and accept on the densified `configs.free` query) gives 90.0 ± 0.7 (5 seeds); a grader-peeking (*exec*) trigger gives 90.3 ± 1.4 (3 seeds)—statistically identical. The audit’s dynamic $\approx$ dense-static equivalence (§A.5) is what makes the honest trigger a faithful proxy; we default to it. Gate density R barely matters: SR $89.5/90.8/91.0$ for $R=1/8/32$.

Budget / time (heavy-tailed, reported honestly). RRT budget $500/2000/6000$ iters $\rightarrow$ SR $89.0/90.2/90.8$. Because the fallback fires only on the hard residual, planning time is *median* 7.1s (mean 9.0s) but with a real tail: p95 ≈ 21 s, max ≈ 27 s (bounded by a 10s RRT timeout). We report the distribution, not just the mean.

Generalization. The fallback lifts SR on every split—global 81.3 ± 2.5 , hybrid 90.0 ± 0.7 , both 89.3 ± 0.2 —and transfers to a second planner (EDMP, 84.5, one seed). The global residual is harder (fewer scenes RRT closes in budget), which we report rather than hide.

Scope caveat. The fallback reaches SOTA-competitive SR *because* these scenes are solvable-by-construction, so a complete planner can rescue the residual. The contribution is the *allocation*—a smooth, dynamically-robust learned+repair plan on the easy majority (Table 2 shows RRT alone is only 67%), with the complete planner reserved for the small hard tail—not a completeness claim on arbitrary or infeasible scenes.

A.8 Positioning vs. published systems

Table 4 places our within-system results next to published numbers. We never compare across the two blocks as a claim: our absolute SR sits ~ 21 pp below the published GPD system (App. A.4 attributes this to the unreleased 7-cost guidance ensemble and trained weights, not to under-training or our repair). The table’s purpose is orientation—GPD+repair is fast and repair-dominated, EDMP is an order of magnitude slower, and classical cuRobo occupies a different speed/quality point.

A.9 Released benchmark and audit protocol

To make the study reusable, we release: (i) the GPU-native GPD/EDMP re-implementation with all ablation configs;

Continuous-feasibility audit (mean $\pm$ std over seeds): waypoint checks over-report success for the compact-Bernstein GPD (17-25pp); the effect is small/n.s. for EDMP. Repair flattens the curve; dotted = dynamic-exec SR.

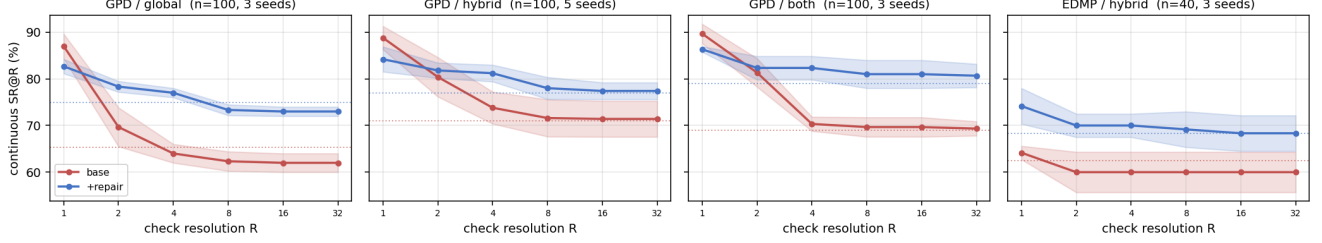


Figure 9: Continuous-feasibility audit across datasets (GPD: global/hybrid/both) and a second planner (EDMP). Base (red) success collapses as the check densifies; repair (blue) is nearly resolution-flat; dotted lines are dynamic-execution SR ($\approx$ the dense static check). The waypoint $\rightarrow$ dense over-reporting is 10–27 pp.

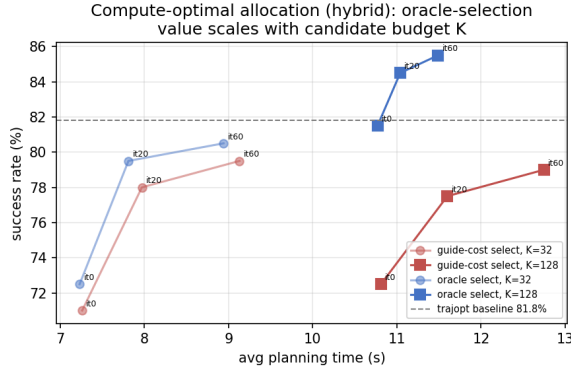


Figure 10: Compute-optimal allocation (hybrid). Oracle (verifier) selection’s benefit grows with candidate budget K , but the 5-seed confirmation of the best point ($K=128$, +trajopt) is 83.5 ± 1.0 — statistically tied with the trajopt baseline (dashed) at higher cost.

(ii) the balanced 400-scene hybrid evaluation set (and the global/both splits) with the continuous-execution success metric; and (iii) the *continuous-feasibility audit protocol*—the $SR@R$ harness (§A.5) that measures how much reported success survives a dense collision check, plus the oracle-gated repair module. The audit is a drop-in evaluation any learned planner can run; we recommend reporting SR at a stated check resolution rather than at waypoints alone, matching classical practice [13].

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Table 4: Positioning. *Top*: our re-implementation (within-system, comparable). *Bottom*: published numbers under each paper’s own protocol (*reference only*; not directly comparable—different collision checks, scene draws, and unreleased weights).

	system	SR%	time
ours	GPD (stitched)	72.9 ± 1.6	7.6s
	GPD + repair	81.8 ± 1.3	9.4s
	GPD + repair + fallback	90.0 ± 0.7	7.1s med.
	EDMP + repair	64.8 ± 1.3	$\sim 51s$
published (ref.)	GPD [2]	92.8	1.9s
	M π Nets [7]	75.8	—
	cuRobo [16]	~ 99	$\sim 0.1s$

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